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Palladium-catalyzed asymmetric propargylic sulfonylation of propargyl esters with sulfonyl hydrazides

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Palladium-catalyzed enantioselective propargylic sulfonylation between racemic propargyl esters and readily available sulfonyl hydrazides with an *N*-benzylamino-bridged bidentate phosphoramidite (*N*-Bn-Me-BIPAM) ligand was reported. The corresponding reactions provided a series of enantiomerically rich propargylic sulfones in good yields with high enantioselectivity under mild reaction conditions.

1. Introduction

Sulfonyl (R-SO_2-R) backbones endowed with distinct electronic and structural features are versatile synthetic intermediates in organic chemistry.[1] As a result, sulfone compounds were widely found in many target molecules of importance in synthetic chemistry, medicinal chemistry, agricultural chemistry, and materials science.[2] The sulfone moiety enhances the efficacy and potency of a drug molecule by forming strong hydrogen bonds with a biological target.[3] Therefore, sulfones not only serve as important building blocks in numerous medicines and drug candidates and natural products, but also play important roles in the discovery of new anticancer agents, antibacterial agents, insecticides and herbicides.[4] In the past few decades, great progress has been made in the synthesis of sulfonyl compounds, and a variety of novel organo- and metal-catalyzed methods have been established.[5] Among these methods, the development of efficient and practical synthetic methods for chiral sulfones has attracted significant interest from synthetic chemists.[6]

Alkynyl sulfones are sulfone derivatives that show very important synthetic utility and are crucial structural moieties that are present in some pharmaceutically and biologically active molecules, such as the inhibitors of cathepsins, cancer Osaka thyroid kinase modulators, DNA cleaving agents, *etc.* (Fig. 1).[7]

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The synthesis of propargyl sulfone backbones has attracted increased attention due to the increase in use,[8] and correspondingly some efficient methods were reported to synthesize the target molecules.[9,10] From 2018, several groups constructed propargyl sulfones under metal-free conditions in good to high yields using methods which utilized propargyl alcohols with sulfinic acids, sodium sulfites or sulfinamides as sulfone sources (Scheme 1a, i–iii).[11–13] Meanwhile, the Lian group reported a palladium-catalyzed multicomponent cross-electrophilic coupling reaction from propargyl acetates, aryl iodides and DABSO to furnish propargyl sulfones in moderate to high yields (Scheme 1b, i).[14] Kleij and Wu reported enantioselective sulfonylation through a copper-catalyzed two-component reaction of propargylic cyclic carbonates with sodium sulfinates or through a three-component reaction of propargylic cyclic carbonates, cyclopropan-1-ols and DABCO·$(SO_2)_2$, allowing the preparation of tertiary terminal propargylic sulfones with quaternary stereocenters in excellent ee values (Scheme 1b, ii).[15] According to previous literature reports,[16] the transition metal-catalyzed decarboxylation of propargylic carbonates delivers either a metal-propargyl intermediate or a metal-allenyl species. Then, the

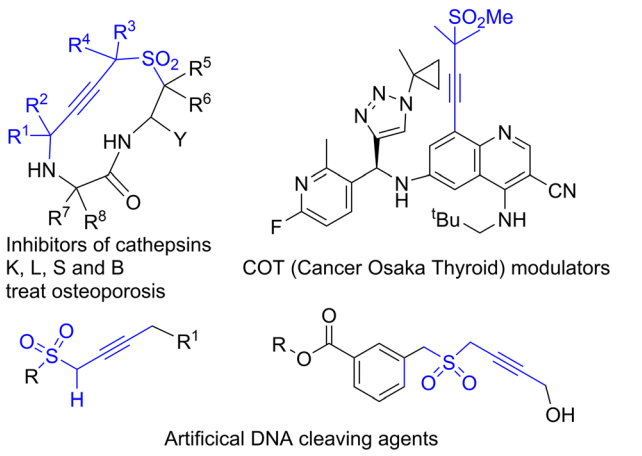
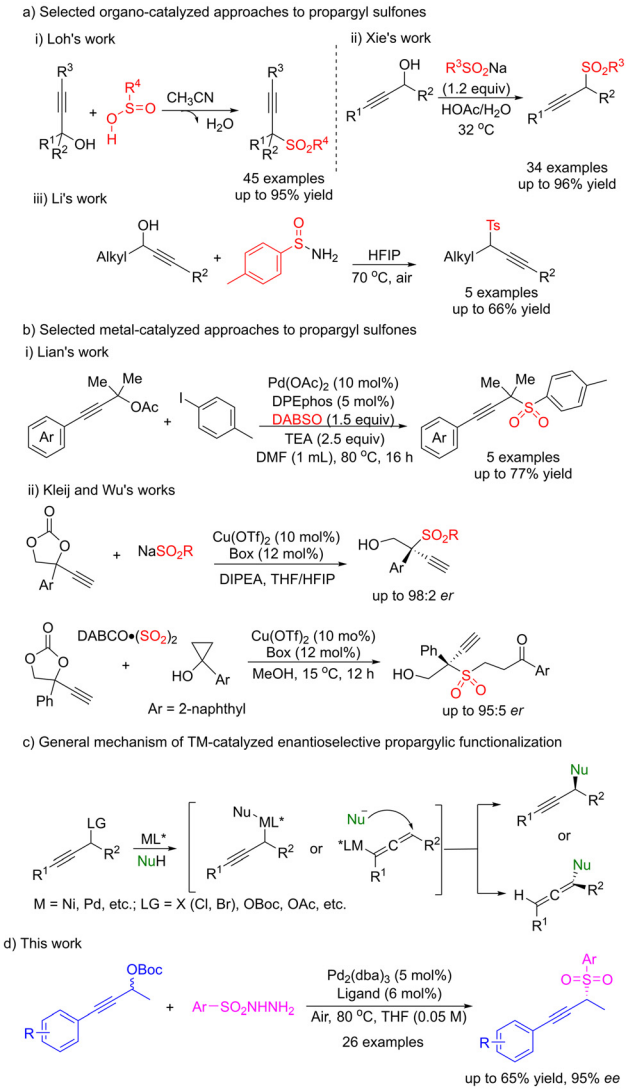


Fig. 1 Bioactive propargyl sulfones derivatives.

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Scheme 1 The synthesis of internal propargyl sulfone compounds.

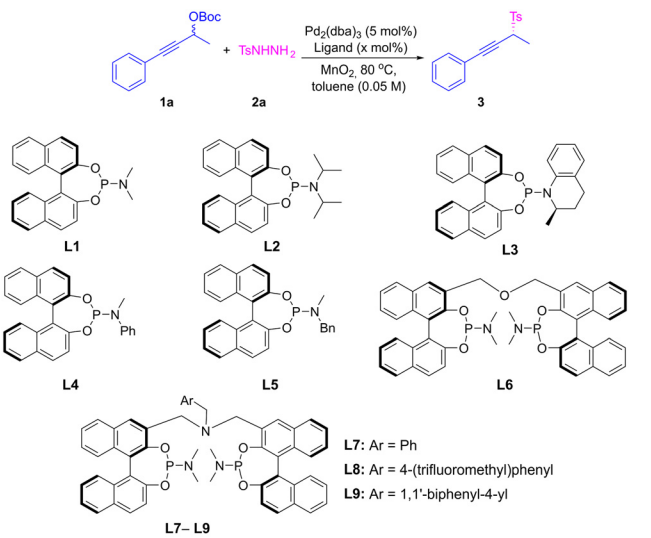
addition of a nucleophile onto a ligand-bound metal-propargyl or metal-allenyl intermediate provides a stereoselective route to access chiral propargyl-functionalized or allenylated products (Scheme 1c).

Herein, we reported a bidentate phosphoramidite-Pd complex-catalyzed asymmetric transformation for the synthesis of chiral propargylic sulfones using propargyl esters with commercially available sulfonyl hydrazides. A series of enantioenriched propargylic sulfones were afforded under mild reaction conditions in moderate yields with good enantioselectivity, highlighting the potential utility of this synthetic method (Scheme 1d).

2. Results and discussion

We commenced the study with the palladium-catalyzed asymmetric propargylic sulfonylation of racemic *tert*-butoxypropargyl ester **1a** with commercially available *p*-toluenesulfonyl hydrazide **2a**. A series of BINOL-derived monodentate and bidentate phosphoramidite ligands **L1–L9** were initially tested with 5 mol% of $Pd_2(dba)_3$ and 5 mol% of MnO_2 as the oxidant. The catalytic results are presented in Table 1 (see SI for full ligand screening results). For monodentate phosphoramidite ligands **L1–L5**, all reactions gave product **3** with less than 58% ee (Table 1, entries 1–5). However, when the bidentate phosphoramidite (Me-Bipam) ligand **L6** was used, the corresponding chiral propargyl sulfone **3** was afforded in 82% ee, albeit in a 17% yield (Table 1, entry 6). Therefore, inspired by this promising result and guided by previous work by Miyaura's group on Me-Bipam ligands,[17] some new chiral bidentate phosphoramidite ligands **L7–L9** bearing bridged benzylamino groups were synthesized. When the ligands **L7–L9** were employed for the reaction, it was found that the product **3** was obtained in 24–55% yield with 54–79% ee values (Table 1, entries 7–9). Of the bidentate phosphoramidite ligands tested, the *N*-Bn-Me-BIPAM ligand **L7** with a bridged benzylamino group turned out to be optimal for this trans-

Table 1 Optimization of the reaction conditions[a]



Entry	Ligand	Yield[b] (%)	ee[c] (%)
1	**L1**	58	50
2	**L2**	49	47
3	**L3**	55	49
4	**L4**	58	27
5	**L5**	53	58
6	**L6**	17	82
7	**L7**	55	79
8	**L8**	24	54
9	**L9**	53	78

[a] Unless otherwise noted, reactions were carried out with **1a** (0.06 mmol), **2a** (0.05 mmol), $Pd_2(dba)_3$ (5 mol%), and MnO_2 (5 mol%) in toluene (1.0 mL) in an oil bath at 80 °C for 12 hours. The ligands **L1–L5** (12 mol%) and the ligands **L6–L9** (6 mol%) were used for the reactions. [b] Isolated yield. [c] Determined using chiral HPLC.

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formation, giving the desired product **3** in 55% yield with a 79% ee value (Table 1, entry 7).

With the optimized chiral ligand **L7** in hand, several other parameters, such as the metal precursor, reaction media, reaction temperature and oxidants, were further explored for the reaction, and the experimental results are summarized in Table 2. Initially, some Pd(0) precursors, including $Pd_2(dba)_3$, $Pd_2(dba)_3{\cdot}CHCl_3$ and $Pd(PPh_3)_4$, were examined, and the corresponding reactions afforded the target compound **3** in 29–55% yield with 53–79% ee values (Table 2, entries 1–3). In contrast, other metal precursors and $Pd(OAc)_2$ showed no reactivity under these reaction conditions (Table 2, entry 4, also see SI for details). Then, several other types of common organic solvents were evaluated for the reaction (Table 2, entries 5–7, see SI for details). THF was the optimal reaction medium for this reaction. Subsequently, the influence of reaction temperature was examined. The reactivity showed a significant temperature dependence, while the enantioselectivity varied modestly (Table 2, entries 5, 8–10). It was found that 80 °C was optimal for the reaction. Different oxidants significantly affected both the reactivity and enantioselectivity (Table 2, entries 11–14). Notably, the use of air as the oxidant afforded the target product **3** in 65% yield with 89% ee value (Table 2, entry 13). Based on its affordability, eco-friendly nature and excellent performance, air was selected as the oxidant for further reactions (see SI for details). When 2,4,6-triisopropylbenzenesulfonyl hydrazide **2b** was employed as a sulfonyl source, the desired product **4** was obtained in 65% yield with a 93% ee value (Table 2, entry 15, see SI for more details). Finally, the optimized reaction conditions were established in the presence of 5.0 mol% of $Pd_2(dba)_3$ and 6.0 mol% of ligand **L7** with air as the oxidant and a substrate ratio of **1a** : **2b** of 1.2 : 1 in THF at 80 °C.

With the optimized reaction conditions established, the substrate scope of various aryl-substituted propargyl esters **1a–1n** in the reaction with **2b** was investigated. The corresponding chiral propargyl sulfones **4–17** were obtained in 47–65% yields with 55–93% ee values. The results are shown in Scheme 2. For the reaction of phenyl-substituted propargyl ester **1a** with **2b**, the product **4** was obtained in 65% yield with a 93% ee value. When the substrates **1b–1d** with –F, –Cl and –Br groups at the *para*-positions of the phenyl rings were investigated for the reaction, the desired products **5–7** were isolated in 47–60% yields, all with 92% ee values. Then, substrates **1e** and **1f** with –Cl and $-OCF_3$ groups at the *meta*-positions of the phenyl rings were utilized for the reaction, and the corresponding products **8** and **9** were delivered in 55% and 47% yields with 78% and 60% ee values, respectively. For substrate **1g** with a *para*-trifluoromethoxy group, the corresponding reaction furnished the desired product **10** in 50% yield with an 80% ee value. On the other hand, substrates **1h–1k** with electron-donating substituents of –OMe, –OEt, –Me, $-^tBu$ at the *para*-positions of the phenyl rings were employed for the reaction and provided the corresponding products **11–14** in 47–54% yields with 55–89% ee values, respectively. For the *ortho*-Me-substituted phenyl propargyl ester **1l**, the reaction produced the desire product **15** in 58% yield with 89% ee. Moreover, substrates **1m** and **1n** disubstituted with methyl groups were also suitable for this reaction, and the corresponding products **16** and **17** were respectively afforded in 61% and 50% yields with 68% and 74% ee

Table 2 Further optimization of the reaction conditions [a]

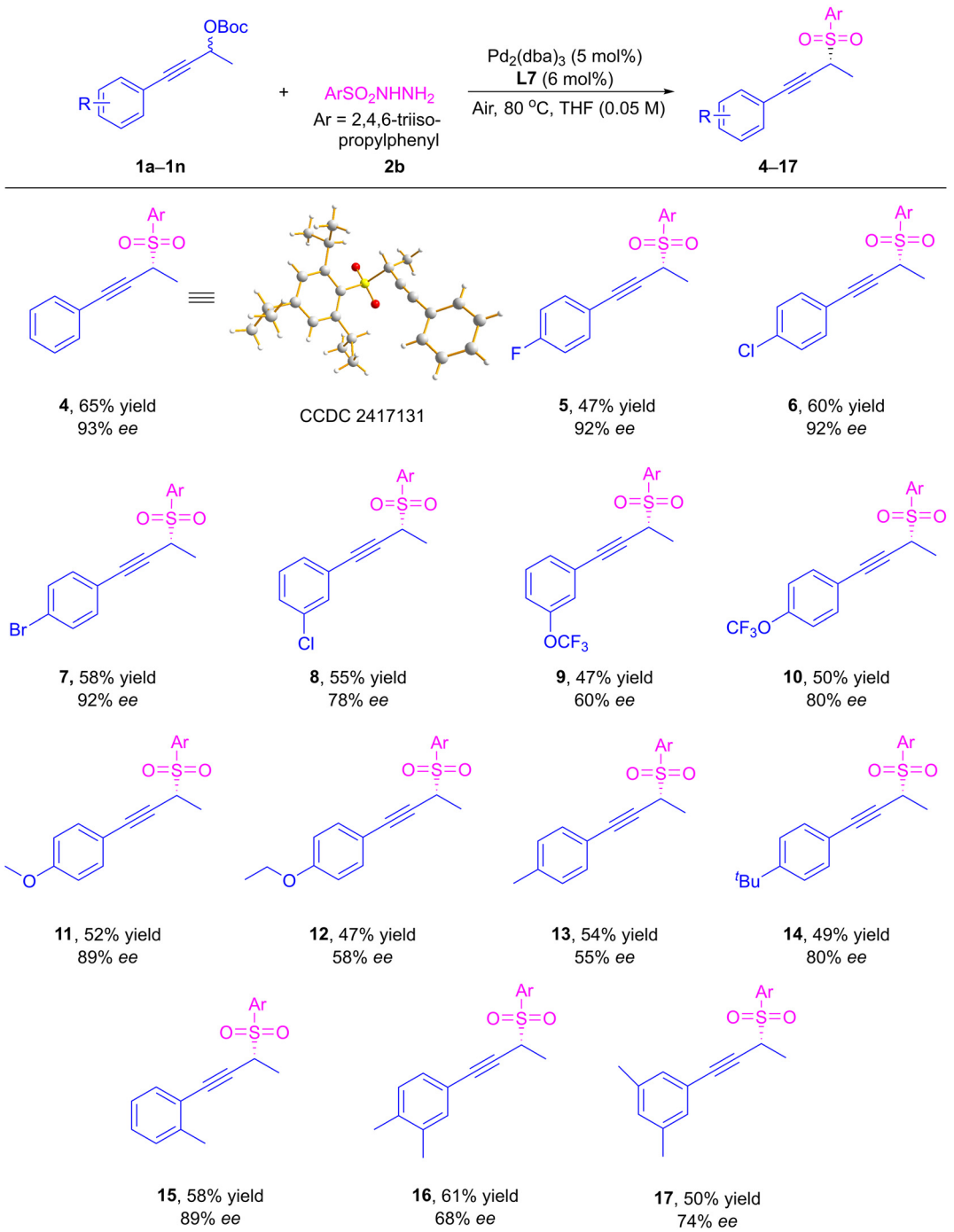


1a **2a**, Ar = *p*-tolyl; **2b**, Ar = 2,4,6-triisopropylphenyl; **3**, Ar = *p*-tolyl; **4**, Ar = 2,4,6-triisopropylphenyl

Entry	Pd	Solvent	Temp. (°C)	Oxidant	Yield[b] (%)	ee[c] (%)
1	$Pd_2(dba)_3$	Toluene	80	MnO_2	55	79
2	$Pd(PPh_3)_4$	Toluene	80	MnO_2	46	75
3	$Pd_2(dba)_3{\cdot}CHCl_3$	Toluene	80	MnO_2	29	53
4	$Pd(OAc)_2$	Toluene	80	MnO_2	nr	—
5	$Pd_2(dba)_3$	THF	80	MnO_2	60	88
6	$Pd_2(dba)_3$	1,4-Dioxane	80	MnO_2	17	84
7	$Pd_2(dba)_3$	DCE	80	MnO_2	44	72
8	$Pd_2(dba)_3$	THF	70	MnO_2	36	78
9	$Pd_2(dba)_3$	THF	90	MnO_2	54	80
10	$Pd_2(dba)_3$	THF	100	MnO_2	49	75
11	$Pd_2(dba)_3$	THF	80	Ag_2O	45	88
12	$Pd_2(dba)_3$	THF	80	$CuSO_4$	20	80
13	$Pd_2(dba)_3$	THF	80	Air	65	89
14	$Pd_2(dba)_3$	THF	80	O_2	58	89
15[d]	$Pd_2(dba)_3$	THF	80	Air	65	93

[a] Unless otherwise noted, reactions were carried out with **1a** (0.06 mmol), **2a** (0.05 mmol), $Pd_2(dba)_3$ (5 mol%), **L7** (6 mol%), and oxidant (5 mol%) in solvent (1.0 mL) in an oil bath at 80 °C for 12 hours. [b] Isolated yield. [c] Determined using chiral HPLC. [d] 2,4,6-Triisopropylbenzenesulfonyl hydrazide.

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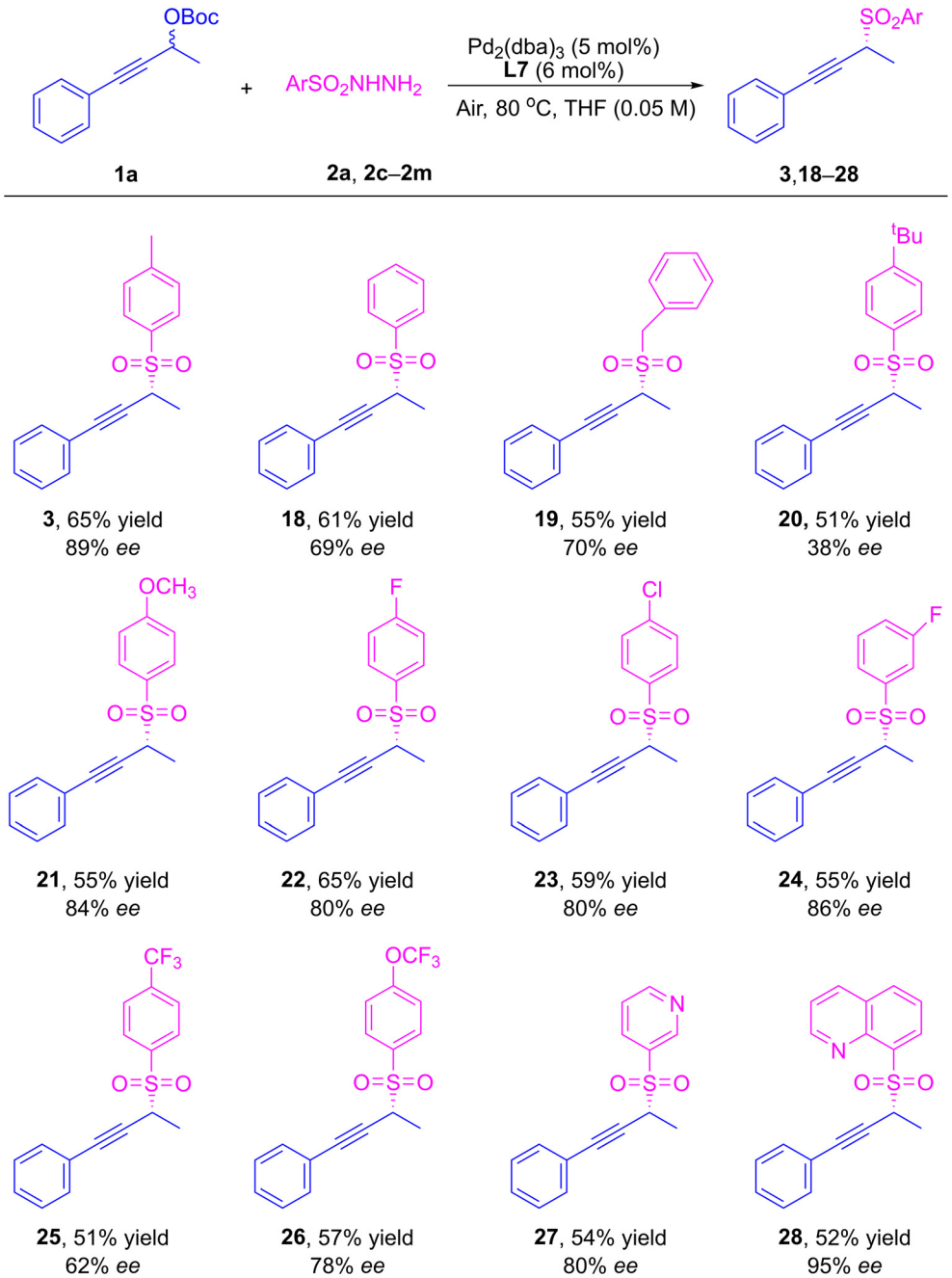
Scheme 2 Substrate scope of the propargyl esters **1**. Reaction conditions: unless otherwise noted, the reactions were carried out with **1a–1n** (0.12 mmol), **2b** (0.10 mmol), $Pd_2(dba)_3$ (5 mol%) and L7 (6 mol%) in an oil bath under an air atmosphere in THF (2.0 mL) at 80 °C for 12 hours.

values. Finally, the absolute configuration of **4**[8] was unambiguously established *via* X-ray crystallographic analysis, while those of the other products were assigned by analogy.

As shown in Scheme 3, the scope of the aryl and heteroaryl sulfonyl hydrazides **2a**, **2c–2m** was examined. The corresponding compounds **3**, **18–28** were synthesized in 51–65% yield with 38–95% ee values. Sulfonyl hydrazides **2a**, **2c** and **2d** with *p*-tolyl, phenyl and benzyl groups were tested first, and the desired products **3**, **18** and **19** were respectively isolated in 55%–65% yields with 69%–89% ee values. Then, sulfonyl hydrazide substrates **2e** and **2f** bearing ^{t}Bu and –OMe groups at the *para*-positions of the phenyl rings were applied for the reaction, and the desired products **20** and **21** were afforded in 51% and 55% yields with 38% and 84% ee values, respectively. When sulfonyl hydrazides **2g–2k** with –F, –Cl, $-CF_3$, $-OCF_3$ groups at the *para*- and *meta*-positions of the phenyl rings

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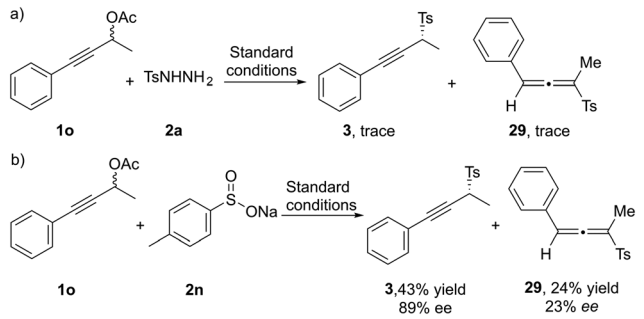


Scheme 3 Substrate scope of the sulfonyl hydrazides **2**. Reaction conditions: unless otherwise noted, the reactions were carried out with **1a** (0.12 mmol), **2a**, **2c–2m** (0.10 mmol), $Pd_2(dba)_3$ (5 mol%) and **L7** (6 mol%) in an oil bath under an air atmosphere in THF (2.0 mL) at 80 °C for 12 hours.

were used, all the reactions proceeded smoothly to provide the desired products **22–26** in 51–65% yields with 62–86% ee values, respectively. Sulfonyl hydrazides **2l** and **2m** with pyridyl and quinolinyl groups were also employed in the reaction, and the corresponding products **27** and **28** were respectively isolated in 54% and 52% yields with 80% and 95% ee values.

Under the optimized reaction conditions, when propargylic acetate **1o** was reacted with *p*-toluenesulfonyl hydrazide **2a**, the corresponding product **3** was only obtained in trace amounts (Scheme 4a). The corresponding results indicate that the $^tBuO^-$ species generated from the Boc-protected propargylic esters after elimination of carbon dioxide (CO_2) likely plays an important role as a base in the subsequent generation of the nucleophilic sulfinate anions (RSO_2^-). Moreover, when propargylic acetate **1o** was treated with sodium *p*-toluenesulfinate **2n**, both propargyl sulfone **3** and allenyl sulfone **29** were obtained in 43% and 24% yields with 89% ee and 23% ee values (Scheme 4b). These results further proved that sulfinate

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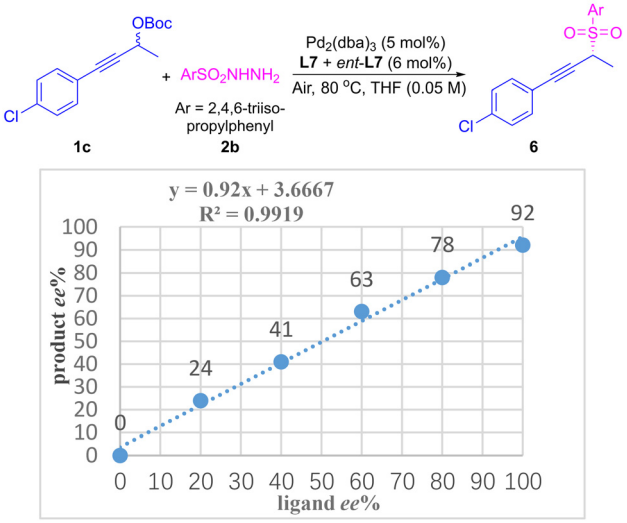


Scheme 4 (a and b) Control experiments.

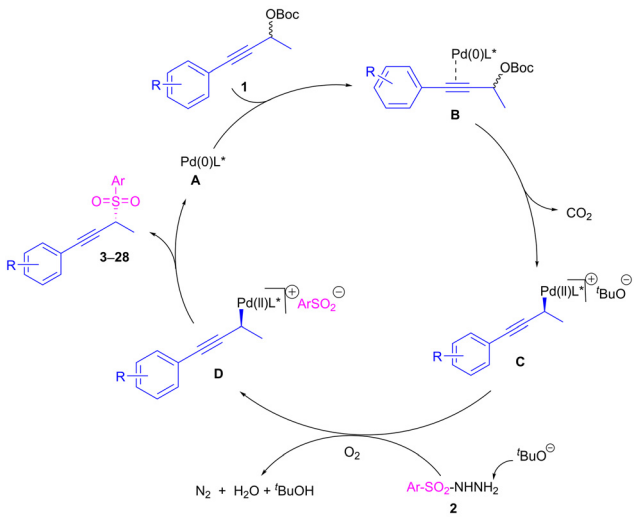
anions are generated from the sulfonyl hydrazides and act as nucleophiles in the reaction.

To gain insight into the possible coordination mode by which the chiral *N*-Bn-Me-BIPAM ligand **L7** and $Pd_2(dba)_3$ form the catalytically active species in the reaction, a nonlinear effects study[18] was carried out on the Pd-catalyzed sulfonylation of propargyl carbonate **1c** with sulfonyl hydrazide **2b** under the optimized reaction conditions. As shown in Scheme 5, the enantiopurity of product **6** displays a linear relationship with the blend of **L7** and *ent*-**L7** that is used, which indicates that a palladium center likely coordinates with a single ligand unit.

On the basis of the control experiments and previous literature reports, we proposed a mechanism[19] for this propargylic sulfonylation, as shown in Scheme 6. Initially, complex **A** of Pd^0L^* is coordinated with the triple bond of propargylic carbonate **1** to form complex **B**, which undergoes oxidative addition with propargylic carbonate **1** to form zwitterionic propargyl Pd^{II} intermediate **C** *via* the elimination of carbon dioxide. The sulfonyl hydrazide **2** is then deprotonated by a *tert*-butoxide ion and is subsequently oxidized to produce the sulfinate anion ($ArSO_2^-$) with the release of N_2 and H_2O. The



Scheme 5 Study of the nonlinear effects.



Scheme 6 The proposed catalytic cycle.

resulting $Pd^{II}L^*$-propargylic sulfinate ($ArSO_2^-$) **D** undergoes a reductive elimination to form the corresponding propargyl sulfone products. Simultaneously, the catalyst is regenerated for the next catalytic cycle.

3. Conclusions

In summary, palladium-catalyzed enantioselective sulfonylation of propargyl carbonates with sulfonyl hydrazides that utilizes a chiral *N*-Bn-Me-BIPAM ligand under mild reaction conditions was developed. This reaction delivered the corresponding propargyl sulfones in good yields with high enantioselectivity. Meanwhile, this transformation has good functional group tolerance, a wide substrate scope, as well as good atom and step economy. It is noteworthy that the employment of air as a promoter ensures the reactivity and delivers optimal catalytic results. Further efforts in our laboratory are focused on expanding the stereoselective transformation of propargyl carbonates into a practical catalytic system.

4. Experimental section

General procedure for the synthesis of the chiral propargyl sulfones.

In a tube, $Pd_2(dba)_3$ (4.5 mg, 0.005 mmol) and **L7** (5.1 mg, 0.006 mmol) were added to anhydrous THF (2.0 ml) under a nitrogen atmosphere. The mixture was stirred for 0.5 hours at room temperature to produce a bright yellow solution. Under an air atmosphere, various substituted propargyl esters (0.12 mmol) and substituted sulfonyl hydrazides (0.10 mmol) were added at 25 °C. Then, the resulting mixture was stirred in an oil bath at 80 °C for 12 hours. After the reaction was complete (monitored by TLC), it was subjected to silica gel column chromatography to afford the desired products.

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Conflicts of interest

There are no conflicts to declare.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request. Experimental details for optimization of reaction conditions, general procedures for the synthesis of the products, characterization of the products, X-ray crystallographic data for the product **4**, and NMR spectra, HPLC analysis of the products can be found in the Supporting Information. See DOI: https://doi.org/10.1039/d5ob01408e.

CCDC 2417131 (**4**) contains the supplementary crystallographic data for this paper.[20]

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